

The Universal Right-Triangle Reflection Sequence (Unequal Legs)

A closed-form rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ and unconditional Class C faithfulness in every dimension

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Abstract

For every right triangle T in the plane with positive **unequal** rational legs, the BFS orbit-growth sequence u_d^T under the group of side-reflections is T -independent — proven through depth 22 and conjectured in general (Theorem 4, Conjecture 5); we call the common sequence the *universal right-triangle reflection sequence*. The translation kernel of the side-reflection representation is computed via Mayer–Vietoris and Crowell as the $\mathbb{Z}[Q]$ -module $I_A \cap I_B$, conjecturally the cyclic ideal $\mathbb{Z}[Q] \cdot h$ on the central generator $h = (c-1)(Y-1)$ (established for the finite quotients $N \leq 16$). In every dimension $n \geq 2$, the right-corner orthoscheme reflection group has closed-form Coxeter–Steinberg growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ (Theorem 16). For pairwise distinct legs the affine representation is unconditionally faithful in every dimension $n \geq 3$ (Theorem 20), via a uniform Schur-complement determinant identity $\det Q_n = -\prod_i a_i^2$ (Lemma 19) together with rank-0 descent on the three Cremona curves 24a1, 32a2, and 72a2. A dimension-independent collision-depth invariant $\text{cd}_n(\lambda) \in \{\infty, 4, 3\}$ determined by the (e, t) -partition statistics of the leg sequence (Theorem 21) classifies first-collision behaviour across all dimensions. The eight length-10 relations, the Schur-complement identity, and the T -independence of the orbit-growth sequence through depth 22 are formally machine-verified in the Lean 4 proof assistant (§7). For the 2D universal sequence the consecutive ratios u_{d+1}/u_d are empirically near 1.50 at the computed depths (descending to ≈ 1.497 at depth 42); the exact asymptotic ratio β_2 , and the algebraic nature of the generating function, remain open and are studied in the companion document `paper_extra.tex`, together with the mod- p state-space and Knuth–Bendix experiments.

1. Setting

Let T be a triangle with vertices $V_0, V_1, V_2 \in \mathbb{Q}^2$. Only the vertex coordinates and the leg lengths a, b are assumed rational. For $i \in \{0, 1, 2\}$ write R_i for the reflection of $\mathbb{R}^2$ across the line containing the side opposite V_i . Set $W(T) = \langle R_0, R_1, R_2 \rangle \leq \text{Isom}(\mathbb{R}^2)$.

When we need explicit coordinates (from §3 onward) we place the right-angle vertex at $B = (a, 0)$ with $A = (0, 0)$, $C = (a, b)$, and write a, b for the two leg lengths.

We explore the orbit of T under $W(T)$ by breadth-first search in the Cayley graph relative to the generating set $\{R_0, R_1, R_2\}$. Let

$$\mathcal{O}_d(T) = \{g \cdot T : g \in W(T) \text{ has minimal word-length } d\},$$

and set $u_d := |\mathcal{O}_d(T)|$, $u_0 = 1$. All distances are computed in exact rational arithmetic and the proofs are checked symbolically over $\mathbb{Q}(a, b)$.

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1.1 Notation

- u_d : BFS layer size at minimal word-length d .
- $\delta_d := F_{d+3} - u_d$: Fibonacci deficit against the Steinberg/Fibonacci baseline.
- $\rho_T : W \rightarrow \text{Aff}(\mathbb{R}^2)$: the affine action of $W = \langle R_0, R_1, R_2 \rangle$ induced by the sides of T .
- $\text{cd}_n(\lambda)$: the collision-depth invariant in dimension n for a leg sequence λ (Theorem 21).
- (e, t) : positional invariant of a leg sequence; e counts endpoint-adjacent equal pairs and t counts triples of consecutive equal legs.

Related work. The n -dimensional rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ derived in §6 is the closed form expected for the growth rate of a path-shaped Coxeter group with one ∞ edge, in the framework of Cannon–Floyd–Parry (Cannon–Wagreich; Floyd–Plotnick). The same family of algebraic integers also appears in birational surface dynamics (Diller–Favre; McMullen; Bedford–Kim); the overlap is at the level of expressions, not of number-theoretic class. Isola–Piergallini (*On the generic triangle group*, 2014–2015) establish that the reflection group of a generic triangle is the expected free product / Coxeter group with no unexpected relations; our unequal-leg right-triangle results are the precedent’s exact rational analogue, replacing genericity by an explicit Diophantine ($\mathbb{Q}$ -rational) faithfulness criterion and adding the closed-form growth data.

2. The Coxeter-group identity (Fibonacci phase)

Assume T is a non-isosceles right triangle with the right angle at V_2 . Then R_0 and R_1 are reflections across the two sides incident to V_2 , meeting at $\pi/2$, hence $R_0 R_1 R_0 R_1 = \text{id}$. The remaining two angles are irrational multiples of π by Niven’s theorem (using the non-isosceles assumption), so no relation $(R_i R_j)^m = 1$ holds for any other pair. The abstract group is

$$W = \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0, 1, 2), (R_0 R_1)^2 = 1 \rangle.$$

By Steinberg’s word-growth formula the Poincaré series is $W(t) = (1+t)^2/(1-t-t^2)$, so $[t^d]W(t) = F_{d+3}$ for $d \geq 1$.

Corollary 1. *For $1 \leq d \leq 9$ (i.e. until the first Euclidean lattice collision at $d = 10$), $u_d = F_{d+3}$. (At $d = 0$, $u_0 = 1 \neq F_3 = 2$.)*

3. Eight length-10 affine relations

Exact-rational breadth-first search yields $u_{10} = 225$, a deficit of 8 from $F_{13} = 233$. These 8 length-10 affine relations are listed in Table 1.

Theorem 2 (Eight length-10 affine relations). *Each of the eight identities of Table 1 holds for every right triangle. With the placement $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$, $a, b > 0$, the affine matrices $M_A = R_{w_A[0]} \cdots R_{w_A[9]}$ and $M_B = R_{w_B[0]} \cdots R_{w_B[9]}$ satisfy $M_A = M_B$ as 3×3 affine matrices over $\mathbb{Q}(a, b)$.*

Proof. This is a finite algebraic verification. Each reflection R_i acts as an affine isometry whose matrix entries are rational functions of a, b over $\mathbb{Q}$ (with denominators dividing a power of $a^2 + b^2$). Each entry of $M_A - M_B$ is a rational function in $\mathbb{Q}(a, b)$; clearing denominators gives a polynomial identity which is verified by SymPy in $\mathbb{Q}(a, b)$ (deterministic, no heuristics). All eight identities have additionally been *formally verified* in the Lean 4 proof assistant: the theorems `rel1_symbolic`, ..., `rel8_symbolic` in `lean/with_mathlib/SymbolicVerification.lean` each discharge the six affine-matrix-entry identities (over $\mathbb{Q}[a, b]$) defining $M_A = M_B$ by the `ring` tactic. As these proofs use only `ring`, they are checked by the Lean kernel, so the identities hold for every right triangle, not merely on a sample of specializations. $\square$ $\square$

#	word A	word B
1	$R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2$	$R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1$
2	$R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2$	$R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1$
3	$R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0$
4	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$
5	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$
6	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$
7	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$
8	$R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1$

Table 1: The eight length-10 affine relations. Each row's two words produce the same affine matrix in the side-reflection action of W on any right triangle.

Proposition 3 (Completeness at depth 10). *Among the 233 canonical Coxeter words of length 10 in W , exactly 225 have pairwise distinct images under ρ_T for every right triangle T — 217 singleton classes and 8 collision pairs, each pair being one of the rows of Table 1. No further length-10 affine identities exist beyond the eight relations and the abstract Coxeter relations.*

Proof. Finite enumeration. All 233 canonical Coxeter words of length 10 are enumerated; the affine matrix of each is computed over $\mathbb{Q}$ on the reference $(3, 4, 5)$ triangle in exact rational arithmetic, the words are grouped by matrix, and the resulting partition has exactly 225 blocks (8 of size 2, 217 of size 1). The eight size-2 blocks are cross-checked against six additional triangles $(5, 12)$, $(8, 15)$, $(7, 24)$, $(1, 2)$, $(2, 3)$, $(1, 7)$, and each block produces the same affine matrix in all seven cases. By Theorem 2 the eight identities hold symbolically over $\mathbb{Q}(a, b)$ and are therefore the only length-10 collisions. $\square$ $\square$

4. Universality across right triangles

By Theorem 2 the kernel N of $W \rightarrow \text{Aff}(\mathbb{R}^2)$ contains the eight length-10 relations in addition to the Coxeter relations, and these relations hold symbolically over $\mathbb{Q}(a, b)$ — so the collision structure they encode is T -independent. We now record precisely what this establishes about u_d^T , and separate it cleanly from the all-depths statement, which remains a conjecture.

Theorem 4 (Universality, established through depth 22). *Let $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1, (R_0 R_1)^2 = 1 \rangle$, and for each right-triangle parameter $T = (a, b)$ with positive rational legs satisfying*

$$a \neq b$$

let $\rho_T : W \rightarrow \text{Aff}(\mathbb{R}^2)$ be the side-reflection realization, with $u_d^T := |\{\rho_T(w) : |w|_W = d\}|$. Then:

- (i) *Every ρ_T has the same linear quotient $W \twoheadrightarrow Q$, determined by the reflection pattern and independent of the legs; hence the translation kernel $T_{\rho_T} = \ker(W \rightarrow Q)$ and its abelianization $T_{\rho_T}^{\text{ab}} \cong I_A \cap I_B$ are the same T -independent $\mathbb{Z}[Q]$ -module for every such T (Lemma 9, Remark 10).*
- (ii) *The eight length-10 relations of Table 1 hold over $\mathbb{Q}(a, b)$ for every such T (Theorem 2); consequently the depth- ≤ 10 collision structure, and in particular u_d^T for $0 \leq d \leq 10$, is independent of T .*
- (iii) *u_d^T is independent of T for all $0 \leq d \leq 22$; this is certified symbolically over $\mathbb{Q}(a, b)$, simultaneously for every right triangle with positive unequal legs, by the Lean 4 theorem `universal_layers_through_22` (Remark 8).*

Conjecture 5 (All-depths universality). *For every right-triangle parameter $T = (a, b)$ with positive rational unequal legs, $\ker(\rho_T)$ is the same subgroup of W , so u_d^T is independent of T for all $d \geq 0$.*

Remark 6 (Status of the all-depths statement). Conjecture 5 is verified through depth 22 (Theorem 4(iii)) and is supported by the T -independent structural facts of (i)–(ii): the common linear quotient Q , the T -independent translation module $I_A \cap I_B$, and the symbolic depth- ≤ 10 relations. These establish that $\ker(\rho_{\text{abs}}) \subseteq \ker(\rho_T)$ for every T (see the factorization below); what is *not* proven in general is the reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\text{abs}})$ uniformly in T , i.e. that no leg-dependent extra collision appears at some depth > 22 . We do not have a complete argument for this and state it honestly as open.

Remark 7 (The isosceles case is genuinely different). The hypothesis $a \neq b$ is essential. When $a = b$, the leg-swap symmetry adds a $\mathbb{Z}/2$ stabilizer to the orbit action, and the orbit sequence becomes 1, 3, 5, 8, 11, 13, 16, $\dots$, which is strictly smaller than the universal sequence $a(n)$ of Table 2 for $n \geq 4$. The universality phenomenon of Theorem 4 does not apply to the isosceles case. See the OEIS entry [A396406](#) and its comments.

Remark 8 (Formal verification of universality through depth 22). The T -independence of the orbit-growth sequence has been *machine-verified* for depths $0 \leq d \leq 22$ in Lean 4 (theorem `universal_layers_through_22` in `lean/with_mathlib/ComputableUniversality.lean`). That proof carries out a single breadth-first sweep over the canonical Coxeter words of W , deduplicating the symbolic affine images $\rho_{(a,b)}(w) \in \text{Aff}(\mathbb{Q}(a, b))$ by their numerators relative to the common denominator $(a^2 + b^2)^{22}$, and certifies that the layer counts are exactly

1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, 351, 554, 875, 1345, 2066, 3203, 4971, 7574, 11543, 17683, 27108, 41067

simultaneously for every right triangle with positive unequal legs. This kernel-checked certificate matches the depth reached by the independent multi-triangle reproduction of §5 and removes any reliance on a single specialization for $d \leq 22$. It establishes Theorem 4(iii); it does *not* establish the all-depths Conjecture 5.

The structural facts behind Theorem 4(i) factor through three lemmas.

Lemma 9 (Mayer–Vietoris reduction). *Let $A = V_4 = \langle R_0, R_1 \mid R_0^2, R_1^2, (R_0 R_1)^2 \rangle$, $B = \mathbb{Z}/2 = \langle R_2 \mid R_2^2 \rangle$, so $W = A * B$. Let $Q = D_\infty \times \mathbb{Z}/2$, with generators X, X', Y, c subject to $X^2 = (X')^2 = Y^2 = c^2 = 1$, $XX' = c$, and c central. Let $T_\rho = \ker(W \twoheadrightarrow Q)$. Then, as left $\mathbb{Z}[Q]$ -modules,*

$$T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B,$$

and this module is independent of T . Setting $h := (c - 1)(Y - 1) \in \mathbb{Z}[Q]$, the cyclic submodule $\mathbb{Z}[Q] \cdot h \subseteq I_A \cap I_B$ satisfies $\mathbb{Z}[Q] \cdot h \cong \mathbb{Z}[Q]/(Y + 1, c + 1) =: M$, since the annihilator of h is generated by $(Y + 1)$ and $(c + 1)$: $(Y + 1)h = 0$ and $(c + 1)h = 0$ in $\mathbb{Z}[Q]$. The equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$ (equivalently $T_\rho^{\text{ab}} \cong M$) is established for the finite quotients $N \leq 16$ and conjectured in general; see the proof. None of the universality results depend on it (Remark 10).

Proof. Since $A \hookrightarrow Q$ and $B \hookrightarrow Q$ are injective, $\mathbb{Z}[Q]$ is free as $\mathbb{Z}[A]$ - and $\mathbb{Z}[B]$ -module, so $H_n(A; \mathbb{Z}[Q]) = H_n(B; \mathbb{Z}[Q]) = 0$ for $n \geq 1$. The Mayer–Vietoris sequence for $W = A * B$ with coefficients $\mathbb{Z}[Q]$ collapses to

$$0 \rightarrow H_1(W; \mathbb{Z}[Q]) \rightarrow \mathbb{Z}[Q]/I_A \oplus \mathbb{Z}[Q]/I_B \rightarrow \mathbb{Z}[Q]/I_W \rightarrow 0,$$

and Crowell’s exact sequence identifies $T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B$ as a left $\mathbb{Z}[Q]$ -submodule.

Membership. $c = XX' \in A$ gives $c - 1 \in I_A$; centrality of c yields $h = (c - 1)(Y - 1) = (Y - 1)(c - 1) \in \mathbb{Z}[Q] \cdot (c - 1) \subseteq I_A$. Likewise $Y - 1 \in I_B$ and $h \in \mathbb{Z}[Q] \cdot (Y - 1) \subseteq I_B$, so $h \in I_A \cap I_B$.

Annihilator. $(Y + 1)(Y - 1) = Y^2 - 1 = 0$ and c central, so $(Y + 1)h = (c - 1)(Y + 1)(Y - 1) = 0$. Similarly $(c + 1)(c - 1) = 0$ gives $(c + 1)h = 0$.

The inclusion $\mathbb{Z}[Q] \cdot h \subseteq I_A \cap I_B$, and evidence for equality. Membership and the annihilator computation give a surjection of left $\mathbb{Z}[Q]$ -modules $\mathbb{Z}[Q]/(Y+1, c+1) \twoheadrightarrow \mathbb{Z}[Q] \cdot h \subseteq I_A \cap I_B$, $1 \mapsto h$, so $\mathbb{Z}[Q] \cdot h \cong M$. We have *not* obtained a complete proof of the reverse inclusion $I_A \cap I_B \subseteq \mathbb{Z}[Q] \cdot h$ over the infinite group Q . What we can certify is the following. Write $Q = D_\infty \times \langle c \rangle$ with $D_\infty = \langle X, Y \rangle$ and let $t = XY$ generate the translation subgroup $\langle t \rangle \cong \mathbb{Z}$. In each finite quotient $D_{2N} \times \langle c \rangle$ (where $t^N = 1$) a direct group-algebra computation gives $\dim(I_A \cap I_B) = \dim(\mathbb{Z}[Q] \cdot h) + 1$ for $3 \leq N \leq 16$, the one-dimensional excess being spanned by the rotation-norm element $\nu_N = \sum_{k=0}^{N-1} t^k$ (verified in `code/ideal/intersection_norm.sage`). The norm ν_N is the standard artifact of the finite cyclic group $\langle t \rangle/t^N$; it has no analogue in the torsion-free $\langle t \rangle \cong \mathbb{Z}$ of the infinite group Q , where $\sum_k t^k$ is not an element of $\mathbb{Z}[Q]$. This is strong evidence that the excess vanishes in Q and hence that $I_A \cap I_B = \mathbb{Z}[Q] \cdot h \cong M$; we record the identification $I_A \cap I_B \cong M$ as established for $N \leq 16$ and conjectural in the infinite limit. Nothing downstream depends on the equality: by Remark 10 the universality results (Theorem 4) use only the T -independence of $I_A \cap I_B$, which is immediate from the Mayer–Vietoris identification above and does not require its cyclic form. $\square$ $\square$

Remark 10 (Universality needs only T -independence, not the explicit M). Theorem 4 requires only that the translation module $T_{\rho_T}^{\text{ab}} \cong I_A \cap I_B$ be *independent of T* . This is manifest from the Mayer–Vietoris identification, whose right-hand side $I_A \cap I_B$ is a purely group-theoretic object attached to $W = A * B \twoheadrightarrow Q$, with no reference to the geometric realization of any triangle. The cyclic submodule $\mathbb{Z}[Q] \cdot h \cong M = \mathbb{Z}[Q]/(Y+1, c+1)$ established above is used only to write the abstract realization ρ_{abs} down concretely (Lemma 13), which needs only the annihilator relations $(Y+1)h = (c+1)h = 0$; neither the universality conclusion nor ρ_{abs} depends on the conjectural full equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$.

Remark 11 (Centrality is essential). The cyclic identification works precisely because $c \in Q$ is central. The naive candidate $(X-1)(Y-1)$ fails because X and Y do not commute in D_∞ . The central element c supplied by the second $\mathbb{Z}/2$ factor of Q unlocks the cyclic presentation, and is intrinsic to the right-triangle structure: the central involution $c = XX'$ realises the half-turn $R_0 R_1$ around the right-angle vertex.

Lemma 12 (Dilation invariance). *For $\lambda > 0$, $\rho_{\lambda T}$ is conjugate to ρ_T by the central dilation D_λ . In particular $\ker(\rho_{\lambda T}) = \ker(\rho_T)$, so the kernel depends only on the similarity class of T , i.e. only on the shape ratio b/a .*

Proof. $D_\lambda \circ \rho_T(R_i) \circ D_\lambda^{-1} = \rho_{\lambda T}(R_i)$ for $i = 0, 1, 2$, since reflecting across a line and then rescaling about the centre is the same as rescaling and reflecting across the scaled line. Conjugation is a group automorphism of $\text{Aff}(\mathbb{R}^2)$, so $\ker(\rho_{\lambda T}) = \ker(\rho_T)$. $\square$ $\square$

Lemma 13 (Abstract realization). *Define $\rho_{\text{abs}} : W \rightarrow Q \ltimes M$ by $R_0 \mapsto (X, 0)$, $R_1 \mapsto (X', 0)$, $R_2 \mapsto (Y, h)$. Then ρ_{abs} is a well-defined group homomorphism.*

Proof. $\rho_{\text{abs}}(R_0)^2 = (X^2, 0) = (1, 0)$ and similarly for R_1 ; $\rho_{\text{abs}}(R_2)^2 = (Y^2, (Y+1)h) = (1, 0)$ by Lemma 9; $((X, 0)(X', 0))^2 = (c^2, 0) = (1, 0)$. $\square$ $\square$

Proof of Theorem 4. (i) Every ρ_T has the same linear quotient $W \twoheadrightarrow Q$, determined by the reflection pattern and not by the leg lengths: the two legs meet at a right angle, so R_0, R_1 map to the two order-2 generators X, X' of the dihedral/central structure of Q , and R_2 maps to Y , uniformly in T . Hence the translation kernel $T_{\rho_T} = \ker(W \rightarrow Q)$ is the same subgroup of W for all T , and by Lemma 9 its abelianization is $T_{\rho_T}^{\text{ab}} \cong I_A \cap I_B$ — a purely group-theoretic module independent of T (Remark 10).

(ii) The eight identities of Table 1 hold as 3×3 affine matrices over $\mathbb{Q}(a, b)$ (Theorem 2), so each is in $\ker(\rho_T)$ for every T ; combined with the abstract Coxeter relations and the depth-10

completeness of Proposition 3, the collision partition of the length- ≤ 10 Coxeter words — and hence u_d^T for $0 \leq d \leq 10$ — is the same for every T .

(iii) This is the content of the Lean 4 certificate `universal_layers_through_22` (Remark 8), which deduplicates the symbolic affine images $\rho_{(a,b)}(w) \in \text{Aff}(\mathbb{Q}(a,b))$ over $\mathbb{Q}(a,b)$ and certifies the layer counts $u_0, \dots, u_{22}$ simultaneously for every right triangle with positive unequal legs.

What this does not prove. By Lemma 13 $\rho_{\text{abs}} : W \rightarrow Q \ltimes M$ is well-defined, and for each T the realization ρ_T factors as $\rho_T = (\text{id}_Q \ltimes \Phi_T) \circ \rho_{\text{abs}}$, where $\Phi_T : M \rightarrow \mathbb{R}^2$ is the $\mathbb{Z}[Q]$ -linear map sending the generator h to the translation $t(\rho_T(R_2))$. This factorization gives the inclusion $\ker(\rho_{\text{abs}}) \subseteq \ker(\rho_T)$ for every T for free. The reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\text{abs}})$ — equivalently, injectivity of Φ_T — would yield $\ker(\rho_T) = \ker(\rho_{\text{abs}})$ for all T and hence all-depths T -independence. We do *not* establish this: $M = \mathbb{Z}[Q]/(Y+1, c+1)$ is free of rank 1 over $\mathbb{Z}[t^{\pm 1}]$ ($t = XY$), hence of infinite rank as an abelian group, and Lemma 12 only relates $\rho_{\lambda T}$ to ρ_T within a single similarity class — it does not compare distinct shapes b/a and does not force Φ_T injective. The all-depths statement is therefore left as Conjecture 5 (see Remark 6). $\square$ $\square$

5. Asymptotic ratio (table of values)

Exact-rational BFS through depth 42 on the reference $(3, 4, 5)$ triangle yields the values in Table 2. The asymptotic behaviour of u_d and the algebraic nature of its generating function are the principal open questions of the program; partial-progress results on conditional transcendence, p -DFAO state-space analysis, and structural recurrence exclusions appear in the companion document `paper_extra.tex`.

d	u_d	ratio	d	u_d	ratio
0	1	—	22	41 067	1.515
1	3	—	23	62 263	1.516
2	5	1.667	24	94 622	1.520
3	8	1.600	25	143 881	1.521
4	13	1.625	26	217 101	1.509
5	21	1.615	27	327 832	1.510
6	34	1.619	28	495 443	1.511
7	55	1.618	29	749 195	1.512
8	89	1.618	30	1 127 236	1.505
9	144	1.618	31	1 697 179	1.506
10	225	1.563	32	2 554 961	1.505
11	351	1.560	33	3 848 384	1.506
12	554	1.578	34	5 777 651	1.501
13	875	1.580	35	8 679 441	1.502
14	1345	1.537	36	13 031 206	1.502
15	2066	1.536	37	19 574 659	1.502
16	3203	1.550	38	29 338 781	1.499
17	4971	1.552	39	43 997 388	1.500
18	7574	1.524	40	65 932 461	1.499
19	11 543	1.524	41	98 849 591	1.499
20	17 683	1.532	42	147 969 934	1.497
21	27 108	1.533			

Table 2: The universal sequence u_d through depth 42.

6. n -dimensional extension: upper bound, closed-form rate, and unconditional Class C faithfulness

The right-triangle analogue in $\mathbb{R}^n$ is the *right-corner orthoscheme*: a simplex with n mutually perpendicular edges meeting at a single vertex. The reflection group W_n is generated by the $n + 1$ face reflections.

6.1 The Coxeter group is independent of legs

Lemma 14 (Perpendicular face pairs, verified symbolically). *For the n -dim right-corner orthoscheme with vertices on the staircase $V_0 = 0$, $V_i = V_{i-1} + a_i e_i$, the perpendicular face pairs are exactly $\{(i, j) : |i - j| \geq 2\}$, independent of the positive leg values $(a_1, \dots, a_n)$. In 3D the dihedral cosines are $\cos \angle(F_0, F_1) = b/\sqrt{a^2 + b^2}$, $\cos \angle(F_1, F_2) = ac/\sqrt{(a^2 + b^2)(b^2 + c^2)}$, $\cos \angle(F_2, F_3) = b/\sqrt{b^2 + c^2}$.*

The abstract Coxeter group is therefore the right-angled Coxeter group with edges (i, j) for $|i - j| \geq 2$ and $m_{ij} = \infty$ for $|i - j| = 1$, the *same group* for every orthoscheme.

6.2 Steinberg upper bound and closed-form r_n

The commutation graph G_n of the n -dim orthoscheme has vertex set $\{0, \dots, n\}$ and edges (i, j) when $|i - j| \geq 2$, with k -clique count $\binom{n-k+2}{k}$. In 3D, Steinberg's formula gives

$$W(t) = \frac{(1+t)^2}{1-2t},$$

with $[t^d]W(t) = 9 \cdot 2^{d-2}$ for $d \geq 2$, $u_0 = 1$, $u_1 = 4$, $u_2 = 9$. Since $u_d \leq [t^d]W(t)$ unconditionally:

Theorem 15 (Unconditional 3D upper bound). *For every right-corner orthoscheme in $\mathbb{R}^3$ with positive legs (a, b, c) , $u_d \leq 9 \cdot 2^{d-2}$ for $d \geq 2$.*

The Coxeter diagram. For pairwise-distinct legs the Coxeter diagram of W_n is a *path* on $n + 1$ nodes (one node per face reflection): by Lemma 14 every non-adjacent face pair $|i - j| \geq 2$ is perpendicular (edge label 2, i.e. no edge drawn), while by Theorem 20 each of the n adjacent pairs $|i - j| = 1$ has a dihedral angle that is an irrational multiple of π , hence generates an infinite dihedral group (edge label ∞). Thus the diagram is the path on $n + 1$ nodes with all n edges labelled ∞ . We do not invoke a closed-form growth rate for this diagram from the literature; instead the rate is computed directly from Steinberg's formula in the proof of Theorem 16 below, and equals the algebraic integer stated there (e.g. $r_2 = \varphi$, $r_3 = 2$, consistent with the $n = 3$ series $(1+t)^2/(1-2t)$).

Theorem 16 (Closed-form n -dim Coxeter rate). *For every $n \geq 2$, the abstract orbit-growth rate of the n -dim right-corner orthoscheme reflection group is*

$$r_n = 1 + 2 \cos\left(\frac{2\pi}{n+3}\right).$$

Proof. By Steinberg's formula applied to G_n , $1/W_n(t^{-1}) = \sum_k (-1)^k \binom{n-k+2}{k} (1+t)^{-k}$. Substituting $y = -1/(1+t)$ collapses the sum to the Fibonacci–Pell polynomial $P_{n+3}(y) = \sum_k \binom{n-k+2}{k} y^k$, defined by $P_0 = 0$, $P_1 = 1$, $P_m = P_{m-1} + yP_{m-2}$. The characteristic equation $x^2 = x + y$ has roots whose ratio $(a/b)^m = 1$ gives $y_j = -1/(4 \cos^2(\pi j/m))$ for $j = 1, \dots, \lfloor (m-1)/2 \rfloor$. Translating back via $y = -1/(1+t)$ and using $4 \cos^2 \theta - 1 = 1 + 2 \cos(2\theta)$:

$$t_j = 1 + 2 \cos\left(\frac{2\pi j}{n+3}\right);$$

the asymptotic rate is the maximum, attained at $j = 1$. □ □

Corollary 17. $3 - r_n = 4 \sin^2(\pi/(n+3)) \sim 4\pi^2/(n+3)^2$, so $r_n \rightarrow 3$.

Remark 18 (Dynamical-degrees connection). The family $\{1 + 2 \cos(2\pi/m)\}_{m \geq 3}$ also arises in birational surface dynamics (McMullen 2007; Bedford–Kim 2009). For $n \neq 3$, r_n is Pisot or totally-real of degree ≥ 3 , hence *not* a Salem number, and by McMullen 2007 Theorem 1.5 cannot be the dynamical degree of a projective rational-surface automorphism.

6.3 Class C unconditional faithfulness

Lemma 19 (Uniform Schur-complement determinant identity). *For every $n \geq 3$ and every positive tuple $(a_1, \dots, a_n)$, the $(n+1) \times (n+1)$ Lorentzian Gram matrix Q_n of the n -dim right-corner orthoscheme satisfies*

$$\det Q_n = - \prod_{i=1}^n a_i^2.$$

Here Q_n is the symmetric tridiagonal matrix with $Q_n[0,0] = 1 - a_1^2$, $Q_n[i,i] = a_i^2 + a_{i+1}^2$ ($1 \leq i \leq n-1$), $Q_n[n,n] = 1$, $Q_n[0,1] = a_2$, $Q_n[i,i+1] = a_i a_{i+2}$ ($1 \leq i \leq n-2$), $Q_n[n-1,n] = a_{n-1}$, read from the staircase normal vectors with the Lorentzian endpoint correction.

Proof. Let $D_k = \det Q_n[:k+1, :k+1]$. Tridiagonal expansion gives $D_k = Q_n[k,k]D_{k-1} - Q_n[k-1,k]^2 D_{k-2}$ with $D_{-1} = 1$. We claim $D_k = P_k := (\prod_{i=1}^k a_i^2)(1 - \sum_{i=1}^{k+1} a_i^2)$ for $1 \leq k \leq n-1$, and $D_n = -\prod_{i=1}^n a_i^2$.

Base. $D_0 = 1 - a_1^2$; $D_1 = (1 - a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2) = P_1$.

Interior step. For $2 \leq k \leq n-1$, the recurrence reduces to the polynomial identity (in the symbols $a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}$ with $S_{k-1} = \sum_{i=1}^{k-1} a_i^2$, $P_{k-2} = \prod_{i=1}^{k-2} a_i^2$)

$$P_{k-2} a_{k-1}^2 a_k^2 (1 - S_{k-1} - a_k^2 - a_{k+1}^2) = (a_k^2 + a_{k+1}^2) P_{k-2} a_{k-1}^2 (1 - S_{k-1} - a_k^2) - (a_{k-1} a_{k+1})^2 P_{k-2} (1 - S_{k-1}),$$

which expands to zero in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$ and is verified symbolically. The identity is independent of k and of n . The general- n determinant identity is thus a complete mathematical proof by the standard tridiagonal cofactor (three-term) expansion together with this induction. The individual *algebraic steps* of the induction — base, $k=2$, inductive, and boundary — are additionally ring-checked in Lean 4 (theorems `schur_base_step`, `schur_k2_step`, `schur_inductive_step`, `schur_boundary_step` in `lean/with_mathlib/SchurGeneral.lean`, each discharged by the `ring` tactic and hence kernel-checked); Lean does *not* prove the general- n determinant identity, which is the mathematical induction above. Separately, the determinant identity $\det Q_n = -\prod_{i=1}^n a_i^2$ is checked by `native_decide` for $n = 2, \dots, 10$ on explicit leg sequences in `lean/RightTriangleReflection.lean`.

Final step. At $k=n$, $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$, and

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2} a_{n-1}^2 [(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})] = - \prod_{i=1}^n a_i^2.$$

The three identities (base, interior, final) close the induction for every $n \geq 3$. □ □

Theorem 20 (Class C unconditional faithfulness, all $n \geq 3$). *For every $n \geq 3$ and every leg tuple $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, the affine representation $\rho : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ of the n -dim right-corner orthoscheme reflection group is faithful, and*

$$u_d(O) = [t^d] W_n(t) \quad \text{for every } d \geq 0.$$

In particular for $n=3$ this is $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

Proof. Write $S = \{R_0, \dots, R_n\}$ for the face reflections and let $m_{ij} \in \{2, 3, \dots, \infty\}$ be the order of $R_i R_j$ in the group $W_n^{\text{geo}} = \langle S \rangle$ actually generated inside $\text{Isom}(\mathbb{R}^n)$. Faithfulness of ρ together with the identity $u_d(O) = [t^d] W_n(t)$ is equivalent to the assertion that (m_{ij}) equals the intended right-angled matrix

$$m_{ij} = 2 \ (|i - j| \geq 2), \quad m_{i,i+1} = \infty \ (0 \leq i \leq n - 1), \quad (*)$$

i.e. that the orthoscheme introduces *no* relation $(R_i R_j)^m = 1$ beyond the perpendicular pairs. We prove $(*)$; the theorem then follows from Tits' theorem applied to the resulting Coxeter system (end of proof). We emphasize that non-degeneracy of Q_n (Lemma 19) does *not* by itself give faithfulness: a non-degenerate Gram form is compatible with a rational dihedral angle. Indeed the isosceles case $a = b$ has $\det Q_n = -\prod a_i^2 \neq 0$ yet $\theta_{01} = \pi/4$, giving the extra relation $(R_0 R_1)^4 = 1$ — the $\text{cd} = 4$ case of Theorem 21. The content of the proof is therefore the angle-irrationality argument that follows; $\det Q_n \neq 0$ enters only to confirm the simplex is non-degenerate for all positive legs.

Step 1 (dihedral angles $\leftrightarrow$ relations). For two faces meeting along a codimension-2 ridge, R_i, R_j generate a finite dihedral group of order $2m$ iff the dihedral angle $\theta_{ij} = \pi/m$ for an integer $m \geq 2$, and an infinite dihedral group (no relation) iff $\theta_{ij} \notin \pi\mathbb{Q}$. By Lemma 14 the pairs with $|i - j| \geq 2$ are perpendicular ($m_{ij} = 2$), giving the right-angled edges of $(*)$. It remains to show every non-perpendicular $\theta_{i,i+1}$ is an irrational multiple of π .

Step 2 (Niven–Conway–Jones constraint). By Lemma 14 each non-perpendicular $\cos^2 \theta_{i,i+1}$ is rational in the legs. If some $\theta_{i,i+1} \in \pi\mathbb{Q}$, then by the extended Niven theorem (Conway–Jones 1976) a rational angle with rational squared cosine satisfies $\cos \theta_{i,i+1} \in \{0, \pm \frac{1}{2}, \pm \frac{\sqrt{2}}{2}, \pm \frac{\sqrt{3}}{2}, \pm 1\}$, i.e. $\cos^2 \theta_{i,i+1} \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. The values 0 (perpendicular) and 1 (degenerate simplex) are excluded, so a genuine extra relation requires $\cos^2 \theta_{i,i+1} \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, i.e. $\theta_{i,i+1} \in \{\pi/3, \pi/4, \pi/6\}$.

Step 3 (boundary dihedrals; elementary). The end edges have $\cos^2 \theta_{01} = b^2/(a^2 + b^2)$ in the local two-leg notation $(a, b) = (a_1, a_2)$, and symmetrically at the far end. Setting this equal to $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ forces $b^2 = \frac{1}{3}a^2, b^2 = a^2, b^2 = 3a^2$ respectively: the first and third give $b/a \in \{1/\sqrt{3}, \sqrt{3}\} \notin \mathbb{Q}$, contradicting rationality of the legs; the middle gives $a = b$, excluded by the pairwise-distinct hypothesis. No boundary dihedral is a rational multiple of π , and no elliptic curve is needed here.

Step 4 (interior dihedrals; square-discriminant $\rightarrow$ Mordell). An interior edge involves three consecutive legs $(a, b, c) = (a_i, a_{i+1}, a_{i+2})$ with b the middle leg, and by Lemma 14

$$\cos^2 \theta_{i,i+1} = \frac{a^2 c^2}{(a^2 + b^2)(b^2 + c^2)}.$$

Setting this equal to a Niven value $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ and clearing denominators yields a quadratic in $B := b^2$. Since the legs are rational, $B = b^2 \in \mathbb{Q}$, so this quadratic has a rational root only if its discriminant is a rational square. The three discriminants are, up to a positive rational square factor, exactly the Diophantine varieties

$$\mathcal{V}_{1/4} : a^4 + 14a^2c^2 + c^4 = m^2, \quad \mathcal{V}_{1/2} : a^4 + 6a^2c^2 + c^4 = k^2, \quad \mathcal{V}_{3/4} : 9a^4 + 30a^2c^2 + 9c^4 = k^2$$

in the two flanking legs (this square-discriminant condition is the correct origin of the three varieties). A rational interior angle thus forces a rational point on one of $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ with $a, c \in \mathbb{Q}_{>0}$. Each is a genus-1 curve in $t = a/c$ whose Jacobian has rank 0, so each has only finitely many rational points:

- $\mathcal{V}_{1/4}$: Jacobian $y^2 = x(x-12)(x-16)$, Cremona 24a1 (conductor 24, rank 0, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); rational points $t = a/c \in \{-1, 0, 1\}$.

- $\mathcal{V}_{1/2}$: Jacobian $y^2 = x^3 - x$, Cremona 32a2 (conductor 32, rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, the four points $(0, 0), (\pm 1, 0), \infty$); rational points $t = 0$.
- $\mathcal{V}_{3/4}$: Jacobian $y^2 = x(x - 300)(x - 1200)$, Cremona 72a2 (conductor 72, rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); rational points $t = 0$.

Every such point gives $ac = 0$ or $a = \pm c$, all excluded by the pairwise-distinct, strictly positive hypothesis on the legs. (Ranks, torsion, Cremona labels, and the rational-point determinations are verified in SageMath in `code/mordell/mordell_24a1.sage`, `mordell_32a2.sage`, `mordell_72a2.sage`, with output committed in the project repository.) Hence no interior dihedral is a rational multiple of π either.

Conclusion. By Steps 3–4 no non-perpendicular $\theta_{i,i+1}$ lies in $\pi\mathbb{Q}$, so $m_{i,i+1} = \infty$, and with Step 1 the geometric Coxeter matrix is exactly $(*)$. Thus W_n^{geo} is the Coxeter system with matrix $(*)$, and Tits’ theorem (Humphreys, *Reflection Groups and Coxeter Groups*, Thm. 5.4) applied to *this established system* gives faithfulness of the geometric representation ρ . Consequently $u_d(O) = \#\{w : \ell(w) = d\} = [t^d] W_n(t)$; for $n = 3$ this is $9 \cdot 2^{d-2}$ for $d \geq 2$. $\square$ $\square$

6.4 The collision-depth invariant $\text{cd}_n \in \{\infty, 4, 3\}$

For a leg sequence $(a_1, \dots, a_n)$ define

$$e := \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0, 1, 2\}, \quad t := \#\{i : 1 \leq i \leq n-2, a_i = a_{i+1} = a_{i+2}\}.$$

e counts *endpoint pairs* (adjacent equal legs at the leftmost or rightmost positions) and t counts *triples*. Let $\text{cd}_n(a_1, \dots, a_n)$ be the minimal word-length at which the orbit BFS first exhibits a non-trivial collision (i.e. the first depth at which $u_d < [t^d] W_n(t)$); set $\text{cd}_n = \infty$ when the affine representation is faithful at every length.

Theorem 21 (Collision-depth invariant, all $n \geq 3$). *For every $n \geq 3$ and every $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$,*

$$\text{cd}_n(a_1, \dots, a_n) = \begin{cases} \infty & \text{if } e = t = 0, \\ 4 & \text{if } e \geq 1, t = 0, \\ 3 & \text{if } t \geq 1. \end{cases}$$

Proof. Faithfulness for $e = t = 0$ is Theorem 20. An endpoint pair $a_i = a_{i+1}$ at a boundary makes the corresponding boundary cosine $b/\sqrt{a^2 + b^2}$ equal $\pm\sqrt{2}/2$, producing the length-4 relation $(R_i R_{i+1})^4 = \text{id}$ and hence $\text{cd} \leq 4$; no shorter relation can appear because $\cos^2 \theta = 0, 1$ are degenerate and the $\pm 1/2, \pm\sqrt{3}/2$ cases reduce as in the proof of Theorem 20 to empty or degenerate loci. An interior triple $a_i = a_{i+1} = a_{i+2}$ makes the middle cosine $ac/\sqrt{(a^2 + b^2)(b^2 + c^2)} = 1/2$ (after the quadratic-in- b^2 analysis with $a = b = c$), producing the length-3 braid relation $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$ and hence $\text{cd} \leq 3$. An interior pair (with no endpoint pair and no triple) leaves all flanking dihedrals irrational multiples of π , by the following elementary argument. For an isolated interior equal pair $a_i = a_{i+1} = a$ with neighbouring leg c ($c \neq a$, else a triple), the relevant dihedral is governed by three consecutive legs (a, a, c) , so by Lemma 14 (with middle leg equal to a)

$$\cos^2 \theta = \frac{a^2 c^2}{(a^2 + a^2)(a^2 + c^2)} = \frac{c^2}{2(a^2 + c^2)},$$

not the generic distinct-leg expression. Setting this equal to the Niven values gives: $\frac{1}{4} \Rightarrow c^2 = a^2$ (i.e. $c = a$, a triple, excluded); $\frac{1}{2} \Rightarrow$ no solution ($c^2 = a^2 + c^2$ is impossible); $\frac{3}{4} \Rightarrow c^2 = -3a^2$, impossible over the reals. The values 0, 1 are degenerate. Hence no flanking dihedral of an isolated interior pair is a rational multiple of π — an elementary computation requiring no elliptic curve — so $\text{cd} = \infty$. (The $\cos^2 \theta = c^2/(2(a^2 + c^2))$ formula and the three Niven specializations are confirmed symbolically in SageMath.) $\square$ $\square$

Note that in $n = 3$ every adjacent equal pair is automatically an endpoint pair, so the formula reduces to the run-length statistic; in $n \geq 4$ *interior* pairs $a_i = a_{i+1}$ (e.g. $a_2 = a_3$ in $(1, 2, 2, 3)$) leave $\text{cd} = \infty$, distinguishing (e, t) sharply from the longest-run statistic.

7. Reproducibility

All computations use exact rational arithmetic; the symbolic verification of Theorem 2 uses SymPy in $\mathbb{Q}(a, b)$, and the Schur-complement identity of Lemma 19 is verified symbolically in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$. The rank-0 descents on the Cremona curves 24a1, 32a2, and 72a2 (for $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ respectively) are verified in SageMath via `EllipticCurve.rank()` and a direct rational-point determination, in `code/mordell1/`. The BFS values in Table 2 are computed by an exact-rational BFS through depth 25 and a Rust disk-streaming BFS through depth 42. Source code and reproduce scripts accompany this paper in the project repository.

Formal (machine-checked) verification. Several of the computational claims are additionally certified in the Lean 4 proof assistant, in the `lean/` subtree of the project repository. A core, Mathlib-free development (`RightTriangleReflection.lean`) machine-checks the basic Coxeter relations, all eight length-10 relations and their explicit affine matrices on the $(3, 4, 5)$ triangle, a direct breadth-first reconstruction of $a(0), \dots, a(17)$, the Fibonacci coincidence $a(n) = F_{n+3}$ for $1 \leq n \leq 9$, and the determinant identity $\det Q_n = -\prod a_i^2$ for $n = 2, \dots, 10$. A second development depending on Mathlib (`with_mathlib/`) proves the eight length-10 relations symbolically over $\mathbb{Q}(a, b)$ (Theorem 2), the algebraic steps of the Schur induction (Lemma 19), and the T -independence of the orbit-growth sequence through depth 22 (Theorem 4(iii), Remark 8), the last holding simultaneously for all right triangles with unequal legs. The all-depths statement (Conjecture 5) is not machine-verified and remains open. The `ring`-based theorems (the symbolic length-10 relations and the Schur induction steps) are verified entirely by the Lean kernel; the remaining results use `native_decide`, whose trusted base additionally includes the Lean compiler and the correctness of the hand-rolled `partial` functions for polynomial determinant and deduplication. Building the projects re-verifies every claim.

On the use of AI

This paper was produced through extended human–AI collaboration, and we state the division of labour plainly. The author is an independent researcher (and AI red-teamer), not a professional mathematician. The mathematical development was AI-led under human direction: the principal model was Anthropic’s Claude, cross-checked against other large language models in a multi-model, adversarial, peer-review-style process that caught and corrected several errors during development. Concretely, the AI did the bulk of the work of deriving and writing the proofs and the text; it wrote the exact-rational breadth-first search and the Rust disk-streaming implementation; it produced the SymPy and SageMath symbolic verifications and the elliptic-curve (rank-0 descent) computations for the three Cremona curves; and it carried out the Lean 4 formalization, including the eight length-10 relations and the Schur induction steps as kernel-checked `ring` proofs and the depth-22 universality certificate via `native_decide`.

The author’s role was to direct the investigation, to choose and pose the problems and the lines of attack, to run the adversarial cross-checking between models, and to verify that the computational artifacts (the BFS code, the SymPy/SageMath scripts, and the Lean developments) actually run and reproduce the stated results. The author takes responsibility for the final text and for any remaining errors. The formal Lean 4 certificates are offered precisely so that the central machine-verifiable claims do not rest on trust in either the human or the AI contributors: they are checked by the Lean kernel (or, where noted, by `native_decide`) and can be re-run

independently.

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